

Smart Authenticity Experience Design With and For Airbnb Hosts

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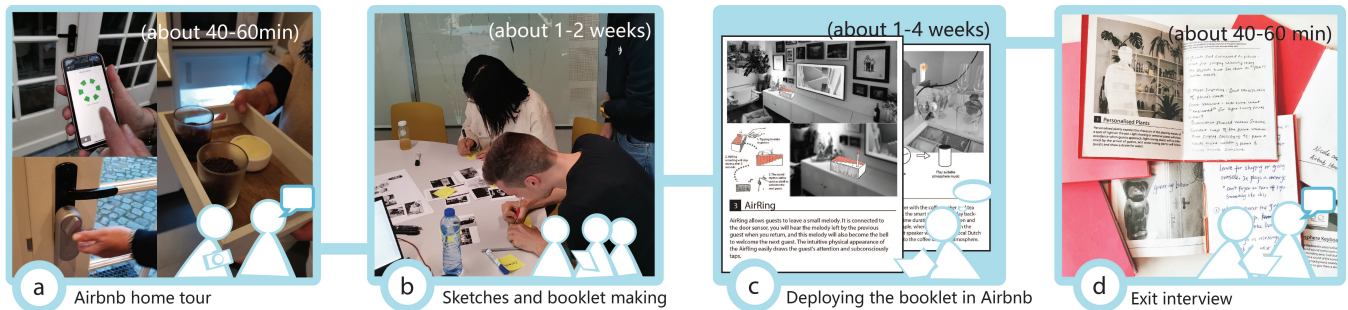


Figure 1: We created bespoke booklet with Airbnb hosts to explore how interaction design can improve guest authenticity: a) We contextually inquired and observed Airbnb hosts' authenticity during the home tour. b) We designed Airbnb's authentic experience around host needs and communicated with them during screening. c) We sketched the bespoke proposal directly onto the host's Airbnb photo and described the product's concept and interaction below then made a booklet and sent it to the host. d) In the exit interview, the hosts explained the booklet notes and reflected on our design.

ABSTRACT

This paper presents a co-design study that aimed to explore how to design authentic experiences for prospective guests by collaborating with seven Airbnb hosts. Through contextual inquiry, we investigated the authentic requirements of different hosts and co-created design ideas with them. Together, we generated 58 interactive design concepts, 19 of which were contributed by the hosts themselves. By discussing these concepts with Airbnb participants and systematically analyzing the qualitative data, we identified five new dimensions for designing authentic guest experiences: serendipity of discovery, sense of welcome, extensions and connections, taking or leaving mementos, and co-generating experiences. Our findings provide valuable insights for designers and researchers looking to create more authentic experiences for Airbnb guests.

CCS CONCEPTS

• **Human-centered computing:** • **Interaction design;** • **Empirical studies in interaction design;**

KEYWORDS

Smart home, Authentic, Guest experience, Co-design, Internet of things, Airbnb

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1 INTRODUCTION

In the hospitality industry, Airbnb has emerged as a leader in offering unique and authentic experiences created by hosts worldwide. With the rise of smart home technologies and the Internet of Things (IoT), many Airbnb hosts have started utilizing these products to enhance the guest experience. For instance, in Hugh11's house, you can just say "Alexa Romance", and the lights will change to a soft color accompanied by selected classical music begins to play [29]. Yet, despite these technological advancements, current IoT applications in Airbnb often fall short in terms of providing authentic experiences. Instead, they focus more on providing basic features for guest convenience and personalized control, such as smart locks, thermostats, motion sensors, and smart lights [8, 22]. These devices are already commonplace in smart hotels like Flyzoo and CitizenM, giving Airbnb a limited advantage in terms of authenticity. The authenticity of Airbnb is mainly embodied in two aspects [33]: The first aspect includes self-making, escape from daily routines, and the pursuit of self-realization, such as autonomy. The second aspect involves bodily feelings, such as sensual pleasures and feelings that are released and consumed.

In recent years, some HCI scholars have begun to focus on the use of IoT in Airbnb. For instance, Papagiannidis and Davlembayeva's [26] survey study concluded that people perceived that smart accommodation can enhance control of stay experiences by improving

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the entertainment experience, aesthetics, and playfulness. They and some other studies (e.g., [8, 22]) also highlighted some negative consequences the new technologies might have to the guests, hosts, and their interactions or relationships. However, to the best of our knowledge, we did not find relevant studies or designs for helping hosts create or amplify the authentic experiences of their offerings. Our research question is: **How can we help hosts to design authentic experiences for prospective guests of an Airbnb?**

In this paper, we present a co-design study conducted with seven Airbnb hosts who reside in the Netherlands. We used contextual inquiry [15] and created bespoke booklets [6] for individual participants (see Fig. 1a). The booklet contained 5-7 imaginary design concepts created by the research team and situated in the host's Airbnb, making it easy for them to evaluate whether they could fit their needs (see Fig. 1b). The hosts reviewed these concepts and created their own designs by revising ours or making new ones (see Fig. 1c). In total, the hosts generated 19 new concepts. We discussed all ideas with them in the exit interview (see Fig. 1d) and then used affinity diagram analysis to cluster the qualitative data and gain insights. As a result, we identified five new dimensions to consider when designing an authentic guest experience: serendipity of discovery, sense of welcome, extensions and connections, take or leave mementos, and co-generate experiences. Compared to previous studies, these dimensions open up a plethora of new possibilities for smart hospitality design. Our study also reveals the tensions and trade-offs the hosts struggle with, between providing optimal guest experiences without limitations and protecting their interests with some controls.

2 BACKGROUND AND RELATED WORKS

2.1 Authentic Experiences in Airbnb Contexts

In the hospitality industry, Airbnb stands out for offering authentic experiences provided by people around the world. Wang's literature study [33] classified authentic experiences into two different aspects. One is the self-making process by which people shape parts of their true selves through travel or adventure. For instance, guests expect to gain a sense of autonomy [17] and self-esteem [21], have their individual requirements considered [17], or experience the locals and neighborhood [19, 27]. The other aspect is bodily feelings, which refer to sensual pleasures, feelings, and other bodily impulses that are released and consumed [33].

From the tourism perspective, authenticity has defined as the level of enjoyment of the tourists and the genuine experiences perceived by tourists [2]. In Airbnb, the homelike sense of belonging allows guests to unwind and feel physically relaxed while also participating in tourist activities such as rehabilitation, recreation, excitement, play [10, 33, 37], or receiving novel experiences [4, 13]. Based on the literature review, our research aimed to clarify the concept of an authentic experience for hosts in simple terms: on the one hand, guests desire a sense of authenticity that includes feeling respected and true to themselves, while also having autonomy and personalized experiences. On the other hand, guests seek to feel at home and experience relaxation and pleasure during their stay in an Airbnb.

To the best of our knowledge, most existing studies on Airbnb's authenticity experience are based on analyzing online reviews or

surveys (i.e., [17–19]), and very little on-site research can be found [3]. However, we have not found any study focused on understanding how hosts create authentic experiences for guests or how they can improve upon them.

2.2 Smart and IoT Devices Used in Airbnbs

IoT devices and smart homes have gained popularity in many countries, with IoT communities and Airbnb forums suggesting commonly used IoT devices in Airbnb such as smart door locks, thermostats, smart lights, and smart speakers, among others [1, 12, 20]. These devices provide guests with flexibility in their check-in and allow them to customize the atmosphere to their liking, although some guests may find the interaction too complicated. To address this issue, some hosts provide guests with guides or attach sticky notes next to the devices [29] to instruct them on proper usage.

Interestingly, some HCI researchers have also identified this gap and are exploring IoT design for Airbnb, but most studies focus only on device usability, privacy, and security [22, 23, 26, 34], which are only partially related to the needs of the authenticity experience. We believe that many unexplored opportunities are waiting to be discovered. Therefore, in this study, we employed qualitative and participatory methods to co-design the authentic experience with and for hosts.

2.3 Co-Design Experiences With and For Hosts

In both the HCI community (e.g., [24, 32, 35]) and in tourism experience research (e.g., [9, 14, 30]), co-design [28] practices involving users or guests have proven to be effective ways to understand their requirements. When designing for a tourism experience, designers can use probes to engage users in creative ways to express their feelings, needs, dreams, and desires, thus enriching the design concepts [30]. However, Desjardins et al. [6] argued that many outputs from co-design studies are too abstract, making it difficult for researchers to further evaluate the rationality of these designs and participants' reflections on them. Building high-fidelity physical prototypes also require a considerable investment of time and money.

To help us quickly start the research and obtain concrete results without compromising quality, we selected the Bespoke Booklet method [5, 6] to create the design probes in our study. This method has been used to explore future smart home products and experiences by investigating inhabitants' needs and sketching ideas on top of photos of their existing environment. It is low-cost and can be tailored specifically to the user's requirements, allowing the design proposal to be personalized for different hosts. We believe it is the best co-design method that fits our research goals.

3 METHOD: CO-DESIGN WITH AIRBNB HOSTS

The bespoke booklets consisted of 10 to 12 situated, imaginary, and bespoke conceptual sketches designed specifically for and with the hosts' Airbnbs. Our co-design was divided into four steps:

3.1 Doing a Contextual Inquiry in a Host's Airbnb

To understand Airbnb and collect contextual information, the host welcomed us and gave us a tour of their Airbnb (see Fig. 1 a). We inquired about Airbnb's characteristics, the strategies employed to improve the guest's experience, and the related content of their interactions with guests during the visit. In addition, together with the host, we took photos and recorded the relevant areas in the rooms involved in the conversation.

3.2 Sketches and Booklet Making

Following the Airbnb tour, our team, consisting of two graduate students and two PhD students majoring in HCI and design, collaborated to generate ideas for potential design concepts based on the unique, special, or inspiring characteristics of each home (see Fig. 1 b). These design concepts were sketched on black-and-white photos of the relevant area in each Airbnb, along with a description of how the designs would interact with guests. During the design process, we used online communication to ensure that the summarized design opportunities aligned with the experience the host wanted to provide. After several rounds of iteration and screening with the host, we selected 5 to 7 bespoke concepts per Airbnb, which were then turned into a paper-based booklet of the same size (3.5 x 5.5 inches). The first section of the booklet contained these designs as well as questions to help the host reflect on them. The second section contained photos provided by the host and a blank part for them to write design notes.

3.3 Deploying the Booklet

To understand if these technologies could support the authentic experience that the host preferred to provide, we distributed the booklets (see Fig. 1 c) to our hosts and asked them to spend some time (approximately 1-4 weeks) reflecting on the concepts designed for their homes and coming up with 5-6 ideas for other designs that could support their guests' authentic experience. We asked them to sketch and describe their own speculative design directly on the following blank pages.

3.4 Exit Interview: Getting the Booklets Back

For a detailed insight into the host's reflections on these bespoke technologies, we met with each host again for a co-design session and an exit interview (40–60 minutes). Based on the booklet, the hosts discussed their reactions as well as new ideas for bespoke interaction designs for authenticity. Meanwhile, we also asked them to share some reflections about the Bespoke Booklets (see Fig. 1 d).

3.5 Study Participants Recruitment

To recruit participants for our study, we used the Airbnb website, Airbnb communities, and our personal networks to find local hosts in the Netherlands. We screened Airbnb based on location, price point, amenities, and room pictures to ensure that these Airbnb features met the needs of the "Authentic travel" guest persona [36] created by Airbnb. We also read host profiles and guest reviews, and we contacted hosts individually to inquire about their willingness to participate. Finally, we chose seven hosts to participate in our

study (see Table 1), and each host spent 2-4 months with us to complete the entire study.

3.6 Data Analysis Method: Affinity Diagramming

We collaborated with hosts to generate 73 design solutions to support the authenticity experience, out of which 58 designs were selected and included in the bespoke booklet (39 from the research team and 19 from the hosts). To analyze the transcribed interviews and bespoke booklet proposals, we used the affinity diagramming method [7, 15]. We divided the transcribed text into sentences, transcribed and outlined the content written and sketched on bespoke by the hosts. We then classified each sentence into several authenticity categories based on its objective. The research team worked collaboratively to classify each sentence, and everyone had to agree on the categorization.

After classifying each sentence, we grouped similar sentences from each authenticity category into broader categories. For instance, if hosts mentioned pets, snacks, and IoT modalities related to welcoming guests, we grouped them into the higher-level category of sense of welcome. Finally, we identified five dimensions of design opportunities to support an authentic guest experience by summarizing the data.

4 RESULTS

We identify five dimensions after analyzing the interview data with affinity diagramming, which will be described in this section.

4.1 Serendipity of Discovery

The hosts wish their Airbnb guests to have fun and unexpected adventures. They use subtle prompts to encourage guests to explore and engage with various elements in a lighthearted and enjoyable way, creating a sense of discovery and surprise, which is the stimulation of a novel experience, a part of bodily feelings in the sense of authenticity. These proposals inspired participants to engage in design thinking, refine the designs, and complete them together. Dorothea said, *"To entice guests to try, I recommend leaving small clues at the base of the candle, such as 'Light me up'."* However, the host is concerned that while serendipity should be part of the guest experience through a lightweight format, too many surprises can make guests feel uncomfortable. In the process of communicating design ideas, Renee had dismissed the idea of a smart water spoon guiding guests to refill the sauna at the right time, saying *"Too much technology... It would be redundant."* Therefore, serendipity-supporting design solutions are suitable to attract guests to actively explore through small cues, and guests can enjoy small surprises to make their journey filled with unexpected adventures. To support lightweight or ephemeral positive surprises that are triggered by guests' autonomous exploration. 19 of 58 bespoke design proposals were related to the serendipity of discovery, in terms of host-recommended experiences (9/19), fun interactions (6/19) and accessibility (4/19): First, hosts recommend novel experiences to guests based on their personal characteristics and observations of guest behavior. Dorothea's Airbnb attracts a lot of young couples, and the candles on the table are used very often, so we proposed the Magic Candle (see Fig. 2 a) to enhance the guests' experience.

Table 1: Overview of hosts and their Airbnb

	Pseudonym	Gender	Airbnb Visiting (average days/year)	Airbnb Hosting (Years)	Home	Location
P1	Bram	Male	15	2	Apartment Inhabited	City
P2	Dorothea	Female	40	5	Idle House	Village
P3	Nicole/Jim	Female/Male	15	4	Apartment Inhabited	City
P4	Renee/William	Female/Male	20	5	Backyard Cottage	Rural area
P5	Mila	Female	65	8	House Inhabited	Rural area
P6	Richard/Lisa	Male/Female	15	2	House Inhabited	City
P7	May/Fisher	Female/Male	30	1	House Inhabited	City

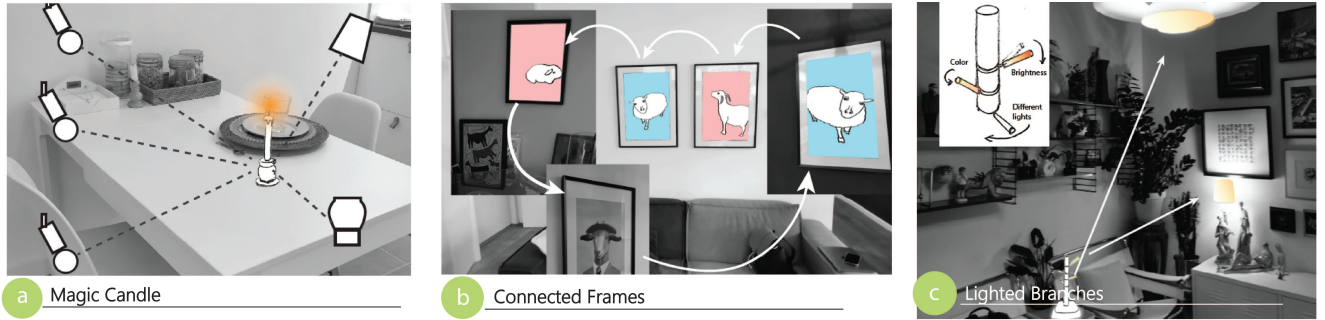


Figure 2: Excerpts from the Bespoke Booklets for the Serendipity of Discovery: a) Idea 4 for P2 Magic Candle conveys the romantic atmosphere Dorothea wants to create, b) Idea 4 for P3 Connected Frames subtly incorporates the mini zoo that Nicole often recommends to her guests into the daily experience, c) Idea 5 for P1 Lighted Branches makes it easy for guests to explore Bram’s complex smart lighting system.

“My daughter exclaimed that it is so romantic.” Secondly, hosts want to use novel technology to make their guests’ stay more interesting. Nicole’s Airbnb has a mini-zoo nearby, and we let the animals inside the zoo interact with guests through frames (see Fig. 2 b): “these frames are fun and will probably make the guests smile:”). Finally, hosts hope to enhance the accessibility of their IoT systems with novel operations. Bram: “The hurdle for guests to experience the devices needs to be lowered”...“Guests will be surprised when they trigger the ambient light while playing with it unconsciously...” (see Fig. 2 c). In general, serendipity can satisfy the need for bodily feelings in authentic experience through novel stimulation.

4.2 Sense of Welcome

The hosts wish to provide a welcoming and comfortable experience for their guests, starting with a warm greeting and ongoing attention to ensure their stay is enjoyable and memorable, which is actually satisfying the guests’ desire for self-esteem, a part of self-making in the sense of authenticity. The context and timing of the welcome are concerns for all hosts. Bram wants his statue bear at the door to greet guests, and these greetings vary in the context of guests check-in, go out, and check-out (see Fig. 3 a): “Every time guests enter or leave the bear, it can have a different greeting”. Nicole focuses on how technology can reinforce the sense of welcome by greeting at more precise times (see Fig. 3 b): “I usually welcome guests with cookies, but nothing beats a really fresh cake from the

oven.” Technology provides the right service at the right time to fulfill the need for authenticity.

To demonstrate hospitality to guests through the act of greeting, a total of 10 designs that use Airbnb items to express welcome to guests appear in five bespoke booklets. The welcome comes in the form of pets (3/10), food (3/10), beautiful physical movements (2/10) or sounds (2/10). Richard’s home has a dog-shaped doorstop, so we created the Welcomed Puppy (see Fig. 3 c). Richard agreed with this solution and expanded on it: “If the puppy can guide guests when they first arrive, it will, even more, surprise them.” Renee decorated the door of her backyard Airbnb with butterflies, which we proposed could greet guests by flapping their wings¹: “Butterflies show hospitality” and “the butterfly’s flapping will surprise the guests.” Some hosts would leave hospitality in the form of wine or snacks, and technology could be more explicit in expressing the intention to “give”. For example, tilting a bowl with snacks to guests or gradually showing them the local food as they approach.

4.3 Extensions and Connections

The authenticity that comes with extension and connection can be summed up in two ways: first, it establishes a quick perception of the guest with the real physical environment, reducing the time it takes for the guest to adapt to the environment, this is the expression of

¹A high-resolution version of each booklet photo and sketch can be found in our supplementary materials in the ACM Digital Library



Figure 3: Excerpts from the Bespoke Booklets for Sense of Welcome and Extension and Connections: a) Idea 6 for P1 Greeting Bear came from Bram's long-term reflection on how guests open the door for different reasons and want the greeting to fit the context. b) Idea 6 for P3 Welcome Cake extends Nicole's habit of leaving cookies for guests. c) Idea 5 for P6 Welcomed Puppy is a doll in Richard's main entrance. d) Idea 7 for P3 Easy Docking lets guests seamlessly connect with Airbnb's items.

autonomy in self-making. Host Nicole designed a dock station to quickly connect the guest's phone with her Wifi and Google Home (see Fig. 3 d): "Now the speakers are connected with Google Home, but I guess nobody is going to be able to get to the speakers... a dock station can do this automatically and play the guest's favorite music". Similarly, Bram has the same motivation: "Guests can quickly get to know the local community". Secondly, technology acts as a mediator to reveal the characteristics of real people. We used the interactive design of Richard's photo wall to share more information about the photos with the guests: "The photo wall is a medium to share my real life, and guests are curious every time." The hosts try to grow and connect so that the guests can relax and feel at home. This helps them feel like they belong, and it also meets the need for bodily feelings in the authentic experience. All of the bespoke booklets involved 16 proposals related to extensions and connections, which included Airbnb devices (6/16), neighborhoods (4/16), hosts (4/16) and other guests (2/16). Bram believes that if guests can easily adjust the Philips wake-up light, it will increase their sense of belonging (see Fig. 4 a): "The guest knows he can set the alarm clock, avoiding annoying conflict between host and guest, it makes guests feel like their own home". Interactive design can also better inform guests about the neighborhood's environment, Renee mentions, "I have guests who come by bike, and some of them don't have any plans, and they need some reference to explore the neighborhood." Finally, hosts also want guests to get to know themselves or other guests better, Dorothea: "I always expect guests to find out that TinTin is my little hint that I'm Belgian" (see Fig. 4 b), Richard: "Technology allows signals to be passed between two guests that no one else can pick up."

4.4 Take or leave Memento

Hosts wish to provide guests with mementos of the local area and encourage them to leave their own mark or memory of their experience in the Airbnb. Five of all design proposals were related to conveying a sense of authenticity through authentic local souvenirs, including items with local characteristics (3/5) or souvenirs generated for/by the guests of the trip (2/5). In the exit interview, Bram recalls from his own travels in Greece, guests can also take home some custom-made specialty items to preserve memories: "The host

will put some rocks from the beach in the Airbnb that guests can decorate with paint and take away." The host's dedication of souvenirs requires the design of a medium, says Richard: "By prompting, guests are surprised to find that these things are welcome to take with them." The hosts not only want guests to take home souvenirs, but they also want guests to leave their own footprints as souvenirs. Renee's room has many paintings left behind by previous guests. Therefore, when she commented on the booklet, she mentioned: "The group photo is a souvenir... Guests can leave these photos representing their memories on Airbnb." (see Fig. 4 c).

4.5 Co-generate Experiences

Hosts wish to facilitate co-created experiences between guests and Airbnb, as well as among past and present guests, in order to foster a sense of community and enhance the overall guest experience. Co-generate refers to the concerted effort of different guests to contribute experience content to Airbnb. Different from the Take or Leave Memento described above, the contribution of experience content is intended to engage more guests. The process of guest generation actually involves guests contributing their own value, which is part of the requirement for self-esteem in authentic experiences. Four proposals contain the idea related to creating an authentic experience by inviting guests to participate in co-creation, including participation in Airbnb life (2/4) and co-creation among different guests (2/4). May and Nicole want guests to feel authentic while co-creating the experience of caring for a pet or plant in Airbnb. May (see Fig. 1): "This cabinet (in the circle) mainly hides the cat's food..." "We can stimulate guests' interest by showing them photos of the cat and teaching them how to feed him properly." Nicole believes that design guides guests to water the plants: "definitely makes it more than a hotel, but a real home :) So it makes it feel more authentic." Bram focuses on the connection that can be made between previous guests: "Guests take pictures of spots they like and leave the pictures for the following guests to reference.", he also wants to leave a corner of the room for guests to create "... Guests can build on the previous guest's creation in one corner."

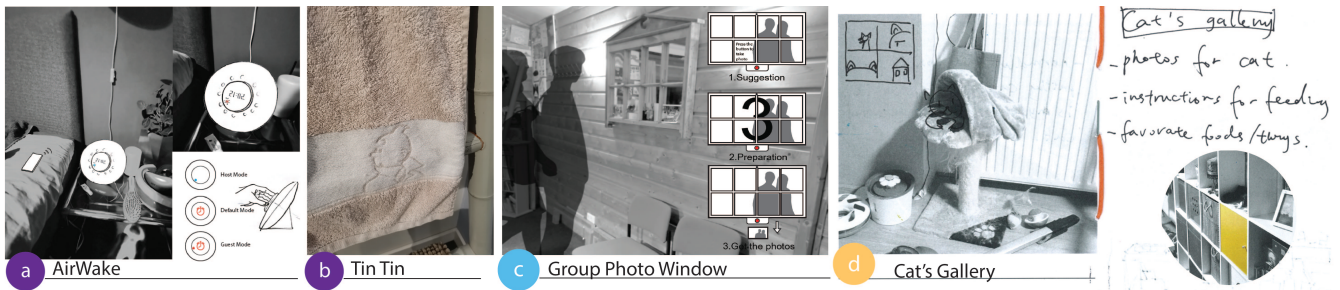


Figure 4: Excerpts from the Bespoke Booklets for Extensions and Connections, Take or Leave Memento and Co-generate Experience: a) Idea 2 for P1 AirWake respects Bram’s daily routine while reducing guest learning costs and increasing guest control over Airbnb. b) Dorothea introduced herself using TinTin. c) Idea 4 for the P4 Group Photo Window matches Renee’s expectation that Airbnb guests leave their mark. d) Idea 5 for P7 Cat’s Gallery was inspired by the Idea 4 Experience Mystery Box (in the circle), which was originally used to store cat food. May wanted to make feeding cats an authentic guest experience.

5 DISCUSSION

Overall, Airbnb hosts expect that interactive designs will support their guests’ authentic experiences in fun and engaging ways. These are often value-added services based on guests’ daily routines. We utilized Desjardins et al. [6, 7] approach to develop our co-design practice. In contrast to previous co-design research in the field of HCI in domestic contexts, the originality of our research is that the theme shifts from the guests’ experience itself to guiding the hosts to create the experience they want to provide, which inevitably confronts tensions between hosts and guests when sharing objects [16], such as access to devices [8], privacy and security [22]. The findings show that we can use interaction design to balance such tensions and propose specific solutions. For Nicole’s bathroom, for example, we created a relaxing atmosphere with ambient lighting and music. Nicole liked the design at first, but she later realized that excessive amusement could lead to more water and electricity use by guests, so she designed her own digital plant, and guests’ energy-saving behavior would be praised and encouraged by the digital plant on the screen. During the co-design process, participants’ comments on the bespoke booklet could help regular hosts understand the study and their role in it, and our design proposals made people think more about it. For example, *“How do we engage guests who have no intention of lighting a candle to try it?”* Dorothea asked about the Magic Candle proposal. *“How can we keep the interactive design from disrupting guests?”* etc. We also found that participants struggled to submit complete design proposals. Renee, for example, was unable to develop her own proposal and was frustrated by a lack of creativity. Often, more came out of the exit interviews after participants talked through their ideas than the actual sketches they made, suggesting that future co-design processes focus on eliciting real experiences and reflections from participants.

Meanwhile, the design proposal should respect the real needs of the host and reduce the interference of the designer’s subjective interpretation, keeping hosts engaged during the design proposal screening is important. Our approach is to provide as many ideas as possible in the early stages and get the initial screening and comments from the hosts, as the design proposals should be fully based on the hosts’ actual requirements. Finally, design practice is easier with the bespoke booklet format. While it does not permit further

reflection through interaction like laying out a real prototype (e.g., [11, 25, 31]), researchers can harvest more diverse thinking and rich data with multiple personalized proposals, and this low-cost approach is ideal for researchers and designers to use in the early stages of exploring user requirements.

Our study has some limitations that should be addressed in future practice. Our participants were all from the Netherlands, and cultural differences may result in different outcomes in terms of how the practice is carried out, such as the hosts’ understanding of the local authenticity experience, and a wider recruitment of participants may yield richer results. However, we thought our participants provided us with quite diverse input. For instance, Dorothea and Bram have a lot of technology in their homes, so they think about the details of the interaction from a technological standpoint; Nicole has environmentalist tendencies, so she covers a lot of green topics.

Many participants liked the concepts we developed together, and some have asked us to realize the idea for their guests. Our next step is to continue the collaborations and make design prototypes for and with the hosts. Two of them have already committed to deploying our designs in their Airbnbs to test whether the prospective guests can perceive the authentic experiences they prepared for them.

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REFERENCES

- [1] Airhostacademy. 2020. Best Smart Home Devices for Airbnb in 2021. <https://airhostacademy.com/best-smart-home-devices-for-airbnb-in-2019/>
- [2] Tugra Nazli Akarsu, Pantea Foroudi, and T. C. Melewar. 2020. What makes Airbnb likeable? Exploring the nexus between service attractiveness, country image, perceived authenticity and experience from a social exchange theory perspective within an emerging economy context. *International Journal of Hospitality Management* 91, August 2019 (2020), 102635. <https://doi.org/10.1016/j.ijhm.2020.102635>

- [3] Jingrui An, Yaliang Chuang, and Pengcheng An. 2021. *Check-In Toolkit for Capturing Guests' Momentary Experiences Without Disturbing Their Traveling*. Springer International Publishing. 581–598 pages. https://doi.org/10.1007/978-3-030-78224-5_40
- [4] Eliane Bucher, Christian Fieseler, Matthes Fleck, and Christoph Lutz. 2018. Authenticity and the Sharing Economy. *Academy of Management Discoveries* 4, 3 (2018), 294–313. <https://doi.org/10.5465/amd.2016.0161>
- [5] Audrey Desjardins, Heidi R. Biggs, Cayla Key, and Jeremy E. Viny. 2020. IoT Data in the Home: Observing Entanglements and Drawing New Encounters. *Conference on Human Factors in Computing Systems - Proceedings* (2020), 1–13. <https://doi.org/10.1145/3313831.3376342>
- [6] Audrey Desjardins, Cayla Key, Heidi R. Biggs, and Kelsey Aschenbeck. 2019. Bespoke booklets: A method for situated co-speculation. *DIS 2019 - Proceedings of the 2019 ACM Designing Interactive Systems Conference* (2019), 697–709. <https://doi.org/10.1145/3322276.3322311>
- [7] Audrey Desjardins, Jeremy E. Viny, Cayla Key, and Nouela Johnston. 2019. Alternative Avenues for IoT: Designing with Non-Stereotypical Homes. In *In CHI Conference on Human Factors in Computing Systems Proceedings*. Glasgow, 1–13. <https://doi.org/10.1145/3290605.3300581>
- [8] Rajib Dey, Sayma Sultana, Afsaneh Razi, and Pamela J. Wisniewski. 2020. Exploring smart home device use by airbnb hosts. In *Conference on Human Factors in Computing Systems - Proceedings*. Association for Computing Machinery. <https://doi.org/10.1145/3334480.3382900>
- [9] Richard Ek, Jonas Larsen, Søren Buhl Hornskov, and Ole Kjær Mansfeldt. 2008. A dynamic framework of tourist experiences: Space-time and performances in the experience economy. *Scandinavian Journal of Hospitality and Tourism* 8, 2 (2008), 122–140. <https://doi.org/10.1080/15022250802110091>
- [10] Cohen Erik. 1988. Authenticity and commoditization in tourism. *Annals of Tourism Research* 15, 3 (1988), 371–386. [https://doi.org/10.1016/0160-7383\(88\)90028-X](https://doi.org/10.1016/0160-7383(88)90028-X)
- [11] William W. Gaver, John Bowers, Kirsten Boehner, Andy Boucher, David W.T. Cameron, Mark Hauenstein, Nadine Jarvis, and Sarah Pennington. 2013. Indoor Weather Stations: Investigating a ludic approach to environmental HCI through batch prototyping. *Conference on Human Factors in Computing Systems - Proceedings* (2013), 3451–3460. <https://doi.org/10.1145/2470654.2466474>
- [12] GuestReady. 2019. Airbnb Smart Homes: Tech devices for hosts. <https://www.guestready.com/blog/airbnb-smart-homes/>
- [13] Daniel Guttentag. 2019. Progress on Airbnb: a literature review. *Journal of Hospitality and Tourism Technology* 10, 3 (2019), 233–263. <https://doi.org/10.1108/JHTT-08-2018-0075>
- [14] Elna Hildén, Kaisa Väänänen, and Pavel Chistov. 2018. Travel experience toolkit: Bus-specific tools for digital service design. *ACM International Conference Proceeding Series* (2018), 193–197. <https://doi.org/10.1145/3282894.3282916>
- [15] Karen Holtzblatt, Jessamyn Burns Wendell, and Shelley Wood. 2005. Building an Affinity Diagram. *Rapid Contextual Design* (2005), 159–179. <https://doi.org/10.1016/b978-012354051-5/50009-4>
- [16] Vinay Koshy and Joon Sung Park. 2021. We just use what they give us: Understanding passenger user perspectives in smart homes. *Conference on Human Factors in Computing Systems - Proceedings* (2021). <https://doi.org/10.1145/3411764.3445598>
- [17] Lidija Lalčić and Christian Weismayer. 2017. The Role of Authenticity in Airbnb Experiences. *Information and Communication Technologies in Tourism 2017* (2017), 781–794. <https://doi.org/10.1007/978-3-319-51168-9>
- [18] Jing Li, Kevin Kam Fung So, and Simon Hudson. 2022. Customer responses to the Airbnb platform: the role of Airbnb experience and memorability. *International Journal of Contemporary Hospitality Management* (2022). <https://doi.org/10.1108/IJCHM-07-2022-0821>
- [19] Lena Jingen Liang, HS Chris Choi, and Marion Joppe. 2018. Understanding repurchase intention of Airbnb consumers: perceived authenticity, electronic word-of-mouth, and price sensitivity. *Journal of Travel and Tourism Marketing* 35, 1 (2018), 73–89. <https://doi.org/10.1080/10548408.2016.1224750>
- [20] David Ludlow. 2021. The best Airbnb smart home tech to keep your guests happy. <https://www.the-ambient.com/guides/best-airbnb-technology-devices-smart-gadgets-2752>
- [21] Paul Lynch. 2017. Mundane welcome: Hospitality as life politics. *Annals of Tourism Research* 64 (2017), 1–12. <https://doi.org/10.1016/j.annals.2017.04.001>
- [22] Shrirang Mare, Franziska Roesner, and Tadayoshi Kohno. 2020. Smart Devices in Airbnbs: Considering Privacy and Security for both Guests and Hosts. *Proceedings on Privacy Enhancing Technologies* 2020, 2 (2020), 436–458. <https://doi.org/10.2478/popets-2020-0035>
- [23] Karola Marky, Sarah Prange, Florian Krell, Max Mühlhäuser, and Florian Alt. 2020. You just can't know about everything: Privacy Perceptions of Smart Home Visitors. *ACM International Conference Proceeding Series* (2020), 83–95. <https://doi.org/10.1145/3428361.3428464>
- [24] Jose Abdelnour Nocera, Daniel Cabrero, Pedro Campos, Frederica Gonçalves, Torkil Clemmensen, Robin Gissing, Lene Nielsen, and Parisa Saadati. 2020. Co-Designing Personas for User Experience and Engagement in Automation. *ACM International Conference Proceeding Series* (2020). <https://doi.org/10.1145/3419249.3420093>
- [25] William Odom, Ron Wakkary, Youn Kyung Lim, Audrey Desjardins, Bart Hengeveld, and Richard Banks. 2016. From research prototype to research product. *Conference on Human Factors in Computing Systems - Proceedings* (2016), 2549–2561. <https://doi.org/10.1145/2858036.2858447>
- [26] Savvas Papagiannidis and Dinara Davlembayeva. 2021. Bringing Smart Home Technology to Peer-to-Peer Accommodation: Exploring the Drivers of Intention to Stay in Smart Accommodation. *Information Systems Frontiers* 0123456789 (2021). <https://doi.org/10.1007/s10796-021-10227-4>
- [27] Dominyka Paulauskaite, Raymond Powell, J. Andres Coca-Stefaniak, and Alastair M. Morrison. 2017. Living like a local: Authentic tourism experiences and the sharing economy. *International Journal of Tourism Research* 19, 6 (2017), 619–628. <https://doi.org/10.1002/jtr.2134>
- [28] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *CoDesign* (2008). <https://doi.org/10.1080/15710880701875068>
- [29] Stephanie. 2020. Could smart technology be useful in an Airbnb listing? <https://community.withairbnb.com/t5/Community-Cafe/Could-smart-technology-be-useful-in-an-Airbnb-listing/td-p/1222523>
- [30] Iis P. Tussyadiah. 2014. Toward a Theoretical Foundation for Experience Design in Tourism. *Journal of Travel Research* 53, 5 (2014), 543–564. <https://doi.org/10.1177/0047287513513172>
- [31] Ron Wakkary, William Odom, Sabrina Hauser, Garnet Hertz, and Henry Lin. 2016. Material speculation: Actual artifacts for critical inquiry. *Interactions* 23, 2 (2016), 44–48. <https://doi.org/10.1145/2889278>
- [32] Ron Wakkary, Doenja Oogjes, Henry W.J. Lin, and Sabrina Hauser. 2018. Philosophers living with the Tilting Bowl. In *Conference on Human Factors in Computing Systems - Proceedings*, Vol. 2018-April. Montreal, 1–12. <https://doi.org/10.1145/3173574.3173668>
- [33] Ning Wang. 1999. Rethinking authenticity in tourism experience. *Annals of Tourism Research* (1999). [https://doi.org/10.1016/S0160-7383\(98\)00103-0](https://doi.org/10.1016/S0160-7383(98)00103-0)
- [34] Huijun Yang, Hanqun Song, Catherine Cheung, and Jieqi Guan. 2021. How to enhance hotel guests' acceptance and experience of smart hotel technology: An examination of visiting intentions. *International Journal of Hospitality Management* 97, April (2021), 103000. <https://doi.org/10.1016/j.ijhm.2021.103000>
- [35] Yaxing Yao, Justin Reed Basdeo, Smirity Kaushik, and Yang Wang. 2019. Defending my castle: A co-design study of privacy mechanisms for smart homes. *Conference on Human Factors in Computing Systems - Proceedings* (2019), 1–12. <https://doi.org/10.1145/3290605.3300428>
- [36] Hui Yi. 2015. A VISIT TO AIRBNB. <https://wordsofasuperheroine.wordpress.com/category/startups/>
- [37] Yunxia Zhu, Mingming Cheng, Jie Wang, Laikun Ma, and Ruochen Jiang. 2019. The construction of home feeling by Airbnb guests in the sharing economy: A semantics perspective. *Annals of Tourism Research* 75, December 2018 (2019), 308–321. <https://doi.org/10.1016/j.annals.2018.12.013>

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